

**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DODOMA REGION
CHAMWINO DISTRICT COUNCIL
FORM FOUR JOINT MID-TERM EXAMINATION
MAKANG'WA, MAKWAWA, MLOWA BARABARANI, MLOWA BWAWANI,
NHINHI, CHIBOLI, MANZASE AND FUFU SECONDARY SCHOOLS.**



032/1

CHEMISTRY 1 MARKING GUIDE

SECTION A (16 MARKS)

1. MULTIPLE CHOICES 10 marks 1@

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	E	C	D	B	A	C	B	C	B

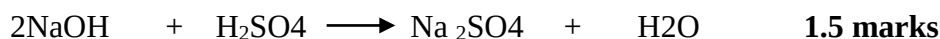
2. MATCHING ITEMS 6 marks 1@

i	ii	iii	iv	v	vi
D	E	B	C	A	F

SECTION B (54 MARKS)

3. (a) Write balanced equation of:

(i) Sodium hydroxide reacts with Sulphuric acid



(ii) Calcium carbonate decomposed by heat

This is a thermal decomposition reaction:

Balanced Equation:



(b) (i) With aid of balanced chemical equations, name the products formed when:

- Potassium nitrate decomposes by heat:

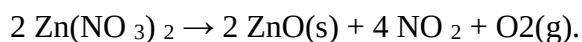
Balanced Equation:



Products: Potassium nitrite and oxygen 1.5 marks

- Zinc nitrate decomposes by heat:

Balanced Equation:



Products: Zinc oxide, nitrogen dioxide, and oxygen 1.5 marks

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(ii) Suggest why nitrates of zinc and potassium behave differently on heating

Answer:

Potassium is an alkali metal, and its nitrates are more thermally stable, so they decompose only partially to give a nitrite and oxygen. **1.5 marks**

Zinc is a transition metal, and its nitrates are less stable; they decompose completely to form the metal oxide, nitrogen dioxide, and oxygen. **1.5 marks**

4. (a) Apparatuses used in volumetric analysis include;

i. Retort stand and clamp

Are used to support and hold the burette in an upright position while performing volumetric analysis experiments.

ii. Burette

It is used to accurately measure and dispense liquids usually acids during titration

iii. Pipette

It is used to measure and transfer definite volume of liquids or solution, usually bases during titration

iv. Measuring cylinder

It is used to measure volumes of liquids

v. Conical flask

It is used to hold solution for titration

(01@ = 5 Marks)

(b) Calculations

i. The balance chemical equations

DATA GIVEN

Mass of Y_2CO_3 is 6.9g

Volume of the solution of Y_2CO_3 is 0.5 L

$V_b = 25\text{cm}^3$

$V_a = 50\text{ cm}^3$

$M_a = 0.1\text{M}$

$M_b = ?$

$n_a = 2$

$n_b = 1$

From,

$M_a V_a n_a = M_b V_b n_b$

$M_b = (M_a V_a n_b) / V_b n_a$

$M_b = (0.1 \times 50 \times 1) / (25 \times 2)$

$M_b = 0.1\text{M}$ 1 Mark

ii. To calculate the relative molecular mass of Y_2CO_3

From,

$\text{Conc (g/dm}^3\text{)} = \text{Molarity} \times \text{molar mass}$

But, $\text{conc} = \text{mass of } Y_2CO_3 / \text{Volume of } Y_2CO_3$.

$\text{Conc. Of } Y_2CO_3 = 6.9 \div 0.5$

$\text{Conc} = 13.9\text{ g/dm}^3$

So, $\text{Molar mass of } Y_2CO_3 = \text{Conc./Molarity}$

$M_r = 13.9/0.1$

$M_r = 139\text{g/mol} = \text{R.M.M} = 139$

Therefore, R.M.M of Y_2CO_3 is 139

1.5 Marks

iii. The R.A.M of Y

$Y_2CO_3 = 139$

$2Y + 12 + (16 \times 3) = 139$

$2Y + 12 + 48 = 139$

$$2Y + 60 = 139$$

$$2Y = 139 - 60.$$

$$2Y = 78$$

$$Y = 39.$$

The name of element Y is potassium (K)

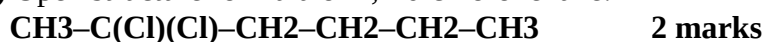
1.5 Marks

5. (a) Five characteristics of a homologous series:

- (i) Same general formula.
- (ii) Gradual change in physical properties.
- (iii) Similar chemical properties.
- (iv) Each member differs by a CH_2 group.
- (v) Same functional group.

5 marks 1@

(b) Open structure formula of 2,2-dichlorohexane:



(c) Why 2,2-dichloro-3-methylbutane is a structural isomer of 2,2-dichloropentane:

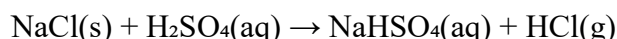
- They have **same molecular formula** but **different structures**. **1 mark**
- They differ in the **arrangement of carbon atoms** (branching). **1 mark**

6. (a) Name of the gas G is Hydrogen chloride (HCl) 1 marks

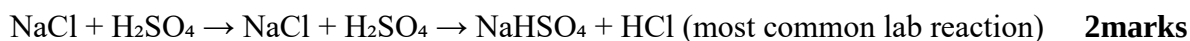
(b) Two chemicals used in the preparation of gas G is:

- (i) Sodium chloride (NaCl) **1 mark**
- (ii) Concentrated sulfuric acid (H_2SO_4) **1 mark**

(c) Write a balanced chemical equation for the reaction of the chemicals named in (b).



(Note: With excess NaCl , another possible reaction is:)

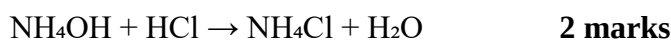


(d) The gas fumes strongly in air, because Hydrogen chloride gas is highly soluble in water. When it comes into contact with the moisture in the air, it forms tiny droplets of hydrochloric acid, which appear as white fumes. 2 marks

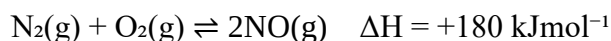
(e) Why does the gas G form dense white fumes with ammonium hydroxide?

Hydrogen chloride reacts with ammonium hydroxide (NH_4OH) to form ammonium chloride (NH_4Cl), a white solid that appears as dense white fumes in moist air.

Reaction:



7. (a) Consider the reversible reaction:



To maximize the yield of NO , let's evaluate each condition:

- (i) Temperature: Use high temperature.

Reason: Since the forward reaction is endothermic (ΔH is positive), increasing the temperature will shift the equilibrium to the right (according to Le Chatelier's Principle), thus favoring the formation of NO and increasing its yield. **2 marks**

(ii) Effect of pressure increase on the rate: There is no significant effect on yield, but rate may increase.

Reason: The number of gas molecules on both sides is equal

(1 mol N₂ + 1 mol O₂ ⇌ 2 mol NO). Therefore, changing pressure has little to no effect on the equilibrium position or yield.

However, increasing pressure generally increases the rate of reaction by increasing the frequency of molecular collisions. **2 marks**

(iii) Effect of adding nitrogen gas (increase concentration): Yes, it will increase NO production.

Reason: Adding more N₂ increases the concentration of one of the reactants.

According to Le Chatelier's Principle, the system will shift to the right to reduce this disturbance, thereby producing more NO. **2 marks**

(b) Three factors determining the production of substances released at electrodes during electrolysis:

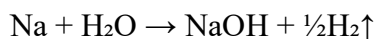
(i) Type of electrolyte – Determines which ions are present in the solution. **1 mark**

(ii) Concentration of ions – Affects which ions are preferentially discharged at the electrodes. **1 mark**

(iii) Nature of electrodes – Inert (e.g., platinum) vs reactive (e.g., copper) electrodes can influence what is deposited or released. **1 mark**

8. (a) Group I elements are called alkali metals because: Group I elements are called alkali metals because they form alkaline solutions when they react with water. For example, sodium (Na) reacts with water to form sodium hydroxide (NaOH), which is a strong base (alkali).

Equation:



The hydroxides formed are soluble in water and increase the pH, making the solution alkaline. **3 marks**

(ii) Because, Non-metals, especially elements like oxygen and halogens, are called oxidizing agents because they have a high tendency to gain electrons during chemical reactions. By gaining electrons, they cause other substances to lose electrons (i.e., get oxidized), while they themselves get reduced.

For example, chlorine (Cl₂) can gain electrons from iron (Fe²⁺), oxidizing iron while being reduced to Cl⁻. **3 marks**

(b) The reaction between dilute hydrochloric acid and sodium thiosulphate results in the production of white precipitates which cause the opaqueness of the solution. The solid substance causing opaqueness is sulphur (S), which forms as a white or yellow precipitate in the reaction:

Equation:



Three uses of sulphur:

(i) Manufacture of sulfuric acid – an important industrial chemical. **1 mark**

(ii) Vulcanization of rubber – to improve elasticity and strength. **1 mark**

(iii) Fungicides and pesticides – used in agriculture to control pests and fungi.

1 marks

SECTION C (30MARKS)

9. Here's how a young scientist might oppose Chabago's argument and support Mgunda's use of manure over fertilizers:

Introduction **2marks**

- (i) Improves Soil Structure: Manure enhances the physical condition of the soil by improving its texture, aeration, and water-holding capacity, which is not provided by chemical fertilizers.
- (ii) Adds Organic Matter: Manure contributes organic matter to the soil, which promotes the growth of beneficial microorganisms and long-term soil fertility, unlike synthetic fertilizers that may degrade soil over time.
- (iii) Reduces Environmental Pollution: Overuse of fertilizers can lead to pollution of water bodies through runoff, causing problems like eutrophication. Manure is more eco-friendly and reduces such risks.
- (iv) Cost-Effective and Locally Available: Manure is often cheaper or free, especially on farms with livestock, making it more sustainable for small-scale farmers compared to costly chemical fertilizers.
- (v) Supplies a Wider Range of Nutrients: While fertilizers mostly provide nitrogen, phosphorus, and potassium, manure contains a broader range of nutrients and trace elements essential for plant growth.

Five points 12.5 marks 2.5@

Conclusion 0.5 marks

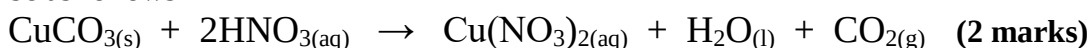
10. (a) X – Hydrated Copper II nitrate $[\text{Cu}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}]$

Y – Nitrogen dioxide gas (NO_2)

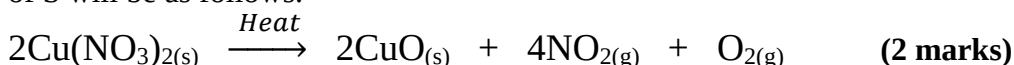
Z – Oxygen gas (O_2)

(2 marks @ = 6 Marks)

(b) Since P is Solid Copper II carbonate (CuCO_3), balanced chemical equation will be as follows



(c) Since Solid S is Copper II Oxide (CuO), the chemical equation for the formation of S will be as follows.



(d) Chemical properties of Z

- i. It supports combustion (relight glowing splint).
- ii. Reacts with metals to form oxides.
- iii. It is strong oxidizing agent.
- iv. Reacts with non-metals to form acidic oxides.

(2 Points @ 1.5 = 3 Marks)

(e) Solid P is Basic copper II carbonate (CuCO_3). **(2 marks)**

11. Introduction **2marks**

Effects of Land Pollution

- (i) Soil Degradation – Pollutants reduce soil fertility, making it less productive for agriculture.
- (ii) Harm to Wildlife – Toxic substances in landfills and waste areas can poison animals and destroy their habitats.

- (iii) Human Health Risks – Exposure to polluted land can cause skin problems, respiratory issues, and other health concerns.

6 marks 2@

Ways of Preventing Land Pollution

- (i) Proper Waste Disposal – Use of landfills, recycling, and composting helps reduce littering and pollution.
- (ii) Use of Eco-friendly Products – Choosing biodegradable and less harmful materials helps reduce chemical pollution.
- (iii) Public Awareness and Education – Teaching communities about pollution and encouraging environmental protection habits can reduce land .

6 marks 1@

Conclusion: **1mark**